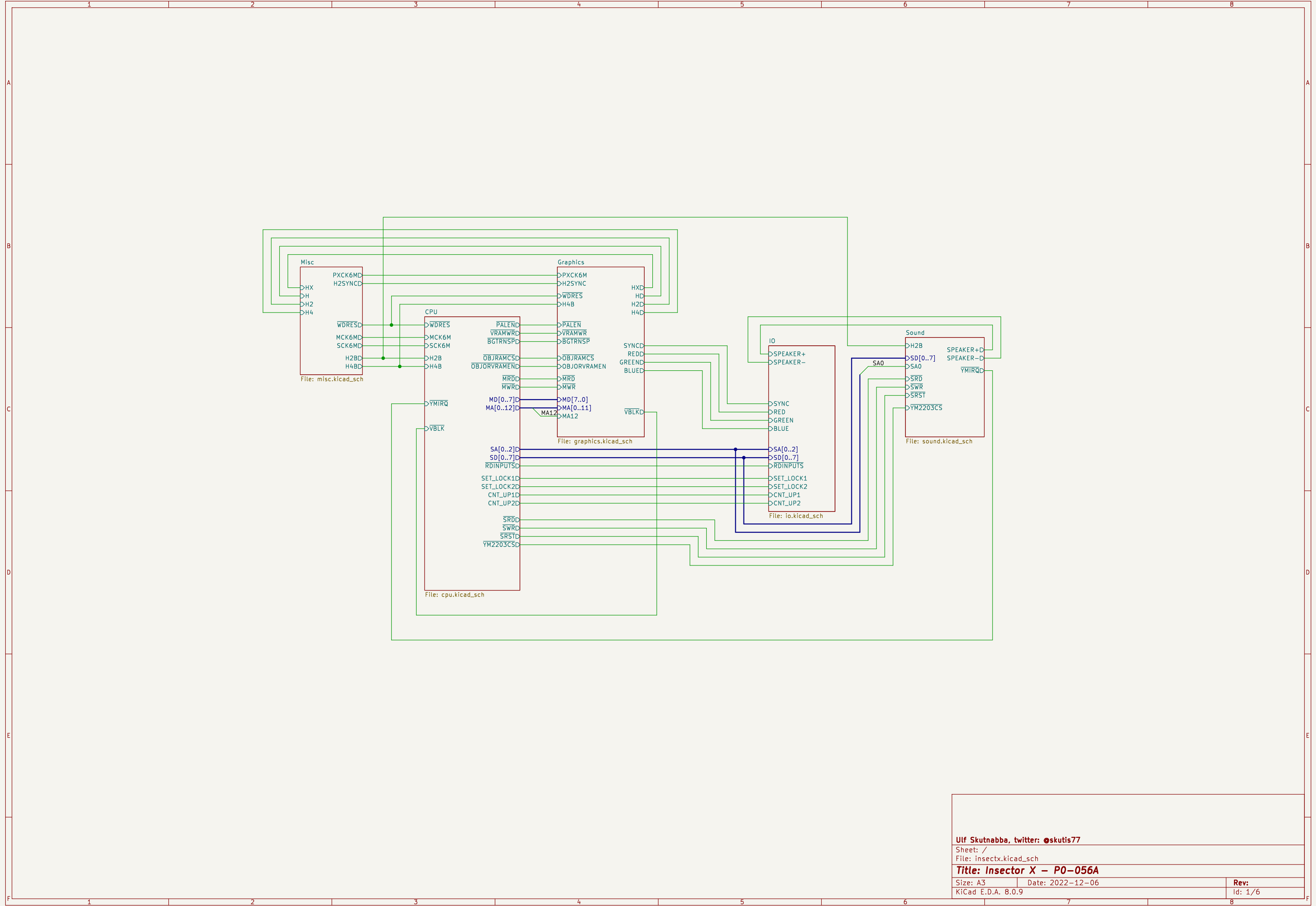




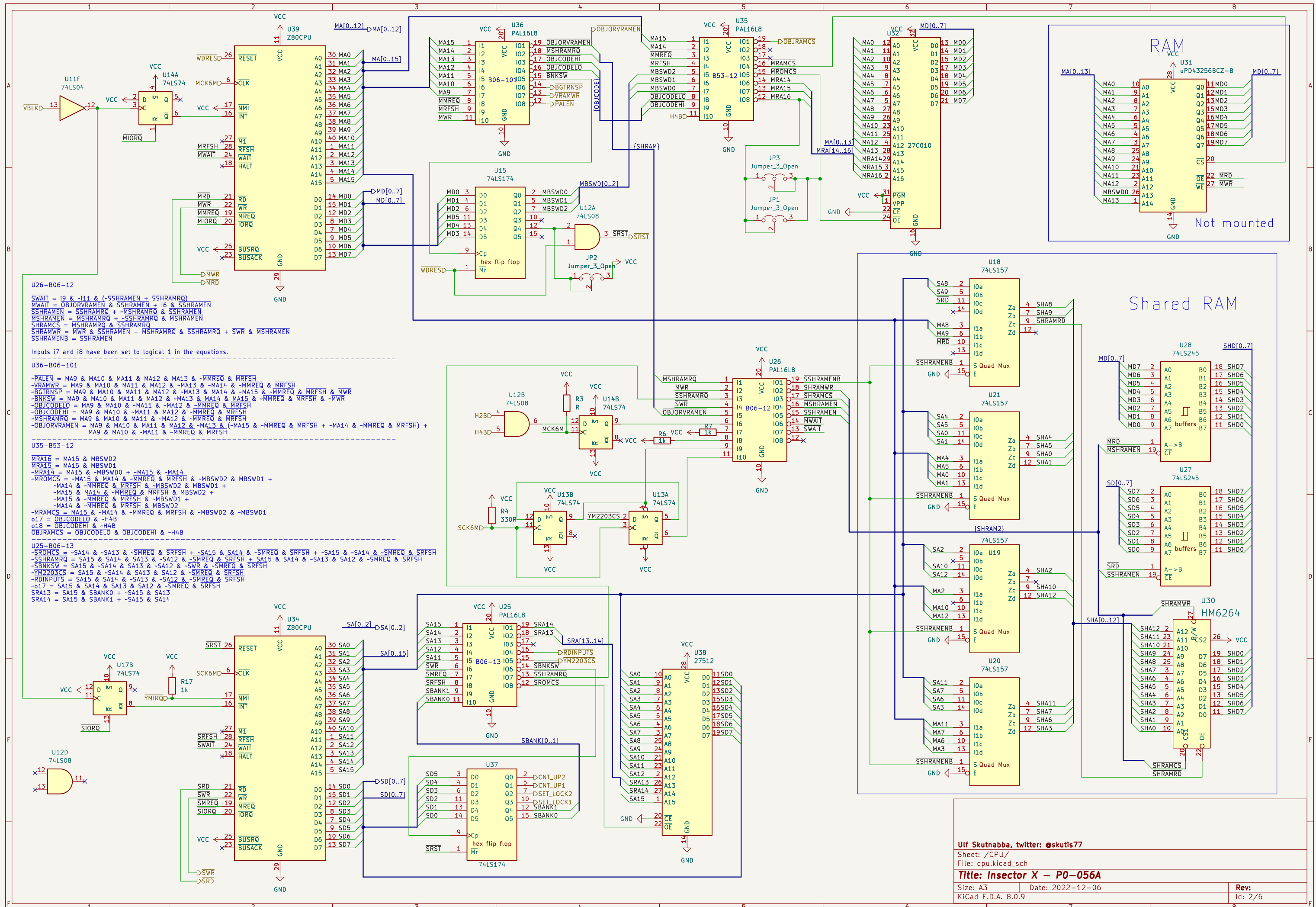
Ulf Skutnabba, twitter: [@skutis77](#)

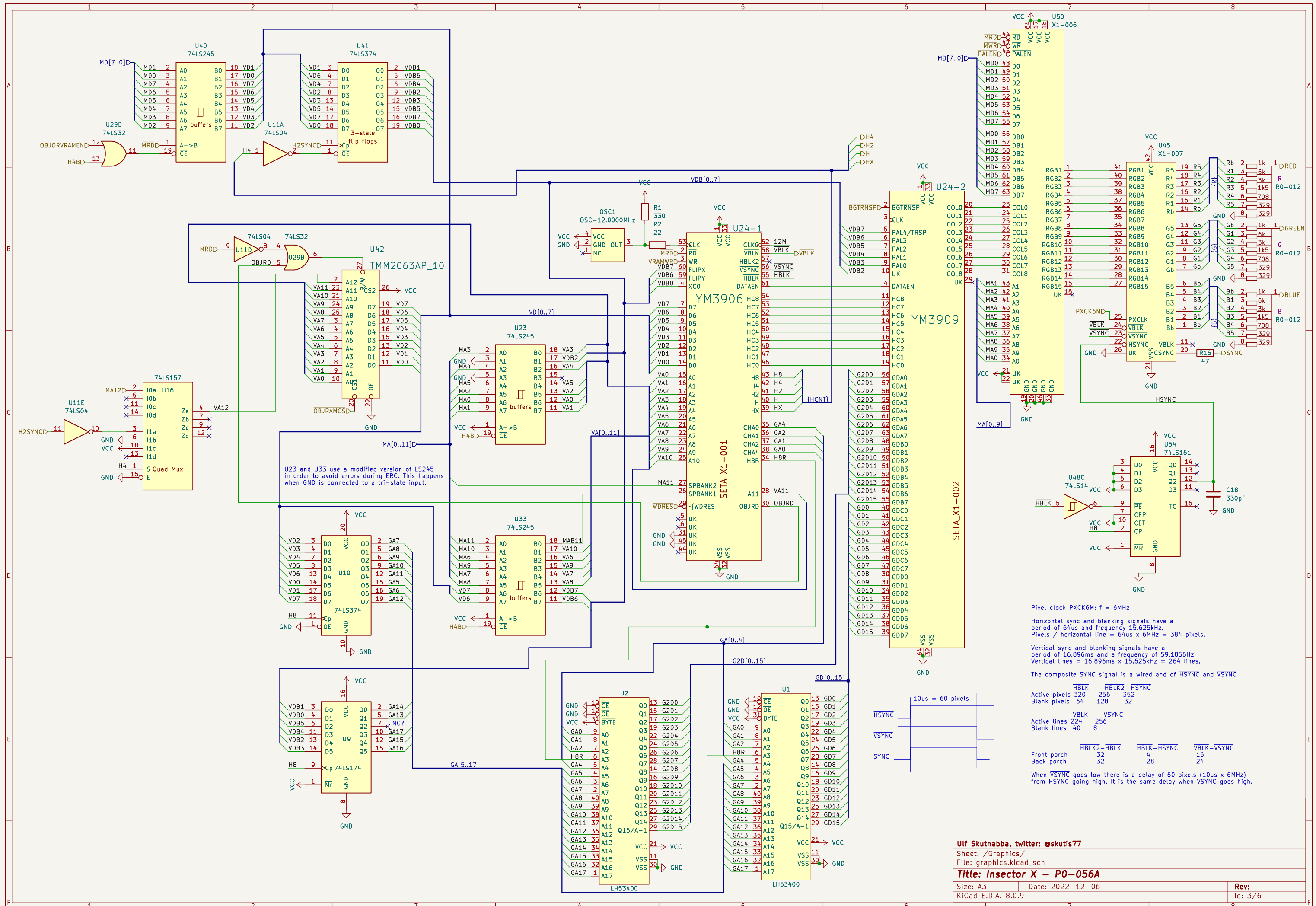
Sheet: /
File: insectx.kicad_sch

Title: Insector X - P0-056A

Size: A3 Date: 2022-12-06
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Pixel clock PXCK6M: $f = 6\text{MHz}$

Horizontal sync and blanking signals have a period of 64us and frequency 15.625kHz. Pixels / horizontal line = $64\text{us} \times 6\text{MHz} = 384$ pixels.

Vertical sync and blanking signals have a period of 16.896ms and a frequency of 59.1856Hz. Vertical lines = $16.896\text{ms} \times 15.625\text{kHz} = 264$ lines.

The composite SYNC signal is a wired and of HSYNC and VSYNC

	HBK1	HBK2	HSYNC
Active pixels	320	256	352
Blank pixels	64	128	32

	VBK1	VSYNC
Active lines	224	256
Blank lines	40	8

	HBK2-HBK1	HBK-HSYNC	VBK-VSYNC
Front porch	32	4	16
Back porch	32	28	24

When VSYNC goes low there is a delay of 60 pixels (10us x 6MHz) from HSYNC going high. It is the same delay when VSYNC goes high.

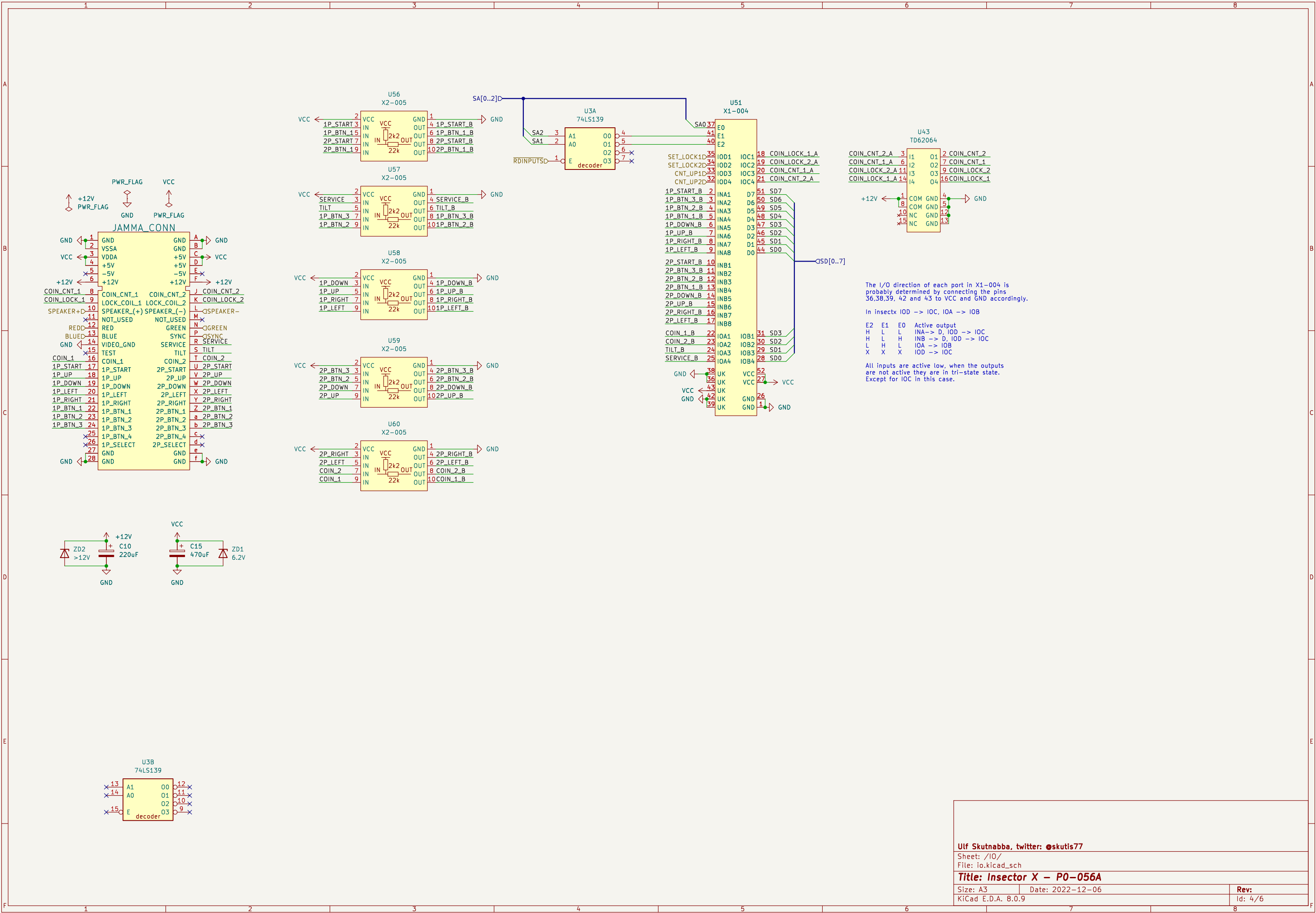
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Title: Insector X - P0-056A

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Date: 2022-12-06
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Sheet: /10/
File: io.kicad_sch

Title: Insector X - P0-056A

Size: A3 Date: 2022-12-06

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Id: 4/6

Rev:
Id: 5/6

